Signal vs System – Continuous LTI property

# Continuous Linear Time-Invariant (LTI) Systems

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# Disc Review

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$

Delta Signal: $\delta[n] = \begin{cases} 1, & \text{if } n=0 \\ 0, & \text{else} \end{cases}$ $\quad \sum_{k=-\infty}^{+\infty} \delta[n] = 1$

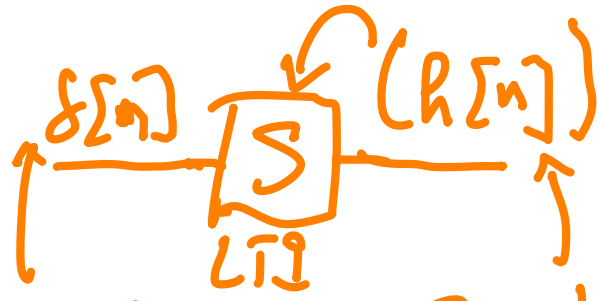
Convolution Computation: 4 step

S1: Pick 1 Fixed Signal (Input Signal)

S2: Reverse Impulse Resp Signal ($\hat{h}[n]$)

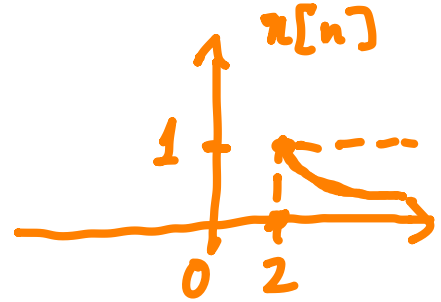
S3: Align Reverse Signal with Fixed Signal and Element wise multiply + Sum all values

S4: Assign value to $y[0]$ and Shift Reverse Signal

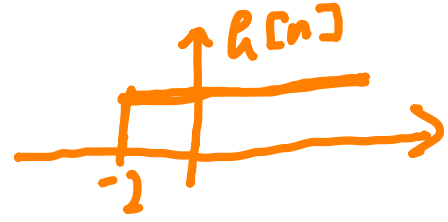


# Ex2.3 Correction

- $x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2]$

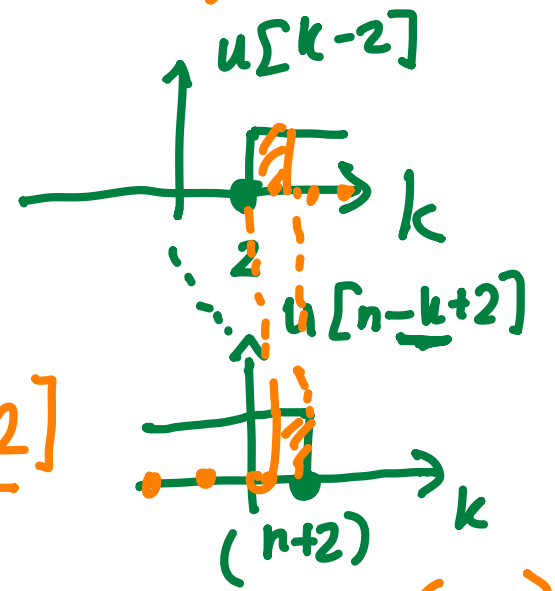



- $h[n] = u[n+2]$




- Compute : $y[n] = x[n] * h[n]$

$$x[n] * y[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$




$$= \sum_{k=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{k-2} u[k-2]\, u[n-k+2]$$

$d = k-2$

$k = d+2$

$$= \sum_{k=2}^{n+2} \left(\frac{1}{2}\right)^{k-2}, \quad n+2 \geq 2 \quad (n \geq 0) \quad (n<0 \Rightarrow y[n]=0)$$

$$= \sum_{d=0}^{n} \left(\frac{1}{2}\right)^{d} = \frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}} = \boxed{2-\left(\frac{1}{2}\right)^{n} = y[n]}$$

- Conclusion: $y[n] = \begin{cases} 2-\left(\frac{1}{2}\right)^n, & n \geq 0 \\ 0, & \text{else} \end{cases}$

Disc LTI Exes:

1. Given: $x[n] = [0\ 4\ \underset{\uparrow}{4}\ 2\ 1\ 3]$, $h[n] = [2\ 1\ \underset{\uparrow}{2}\ 2]$ ⇒ Compute Conv

- $\hat{h}[n] = h[-n] = [2\ 2\ \underset{\uparrow}{1}\ 2]$

- $y[0] = [0\ 4\ 4\ 2\ 1\ 3] = 8 + 8 + 2 + 2 = 20$

  $[2\ 2\ 1\ 2]$

- $y[-1] = [0\ 4\ 4\ 2\ 1\ 3] = 0 + 8 + 4 + 4 = 16$

  $[2\ 2\ 1\ 2]$

- $y[-2] = [0\ 4\ 4\ 2\ 1\ 3] = 0 + 0 + 4 + 8 = 12$

  $[2\ 2\ 1\ 2]$

- $y[-3] = [0\ 4\ 4\ 2\ 1\ 3] = 8$

$[2\ 2\ 1\ 2\ 0]$

~~$y[-1] = [0\ 4\ 4\ 2\ 1\ 3] = 8+8+2+2 = 20$~~

~~$[2\ 2\ 1\ 2]$~~

- $y[1] = [0\ 4\ 4\ 2\ 1\ 3] = 8+4+1+6 = 19$

$[2\ 2\ 1\ 2]$

- $y[1] = [0\ 4\ 4\ 2\ 1\ 3] = 4+2+3 = 9$

$[0\ 2\ 2\ 1\ 2]$

- $y[3] = [0\ 4\ 4\ 2\ 1\ 3] = 2+6 = 8$

$[0\ 0\ 2\ 2\ 1\ 2]$

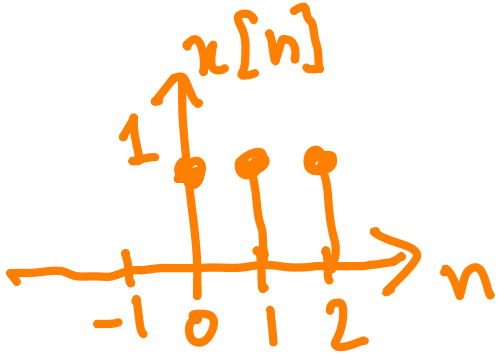
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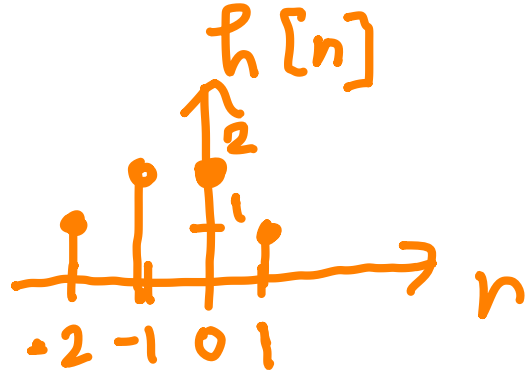
- $y[4] = [0 \quad 4 \quad 4 \quad 2 \mid 3] = 6$

$[0 \quad 0 \quad 0 \quad 2 \quad 2 \mid 2]$

- Conclusion : $y[n] = [8, 12, 16, 20, 19, 9, 8, 6]$

2. Given:




Compute and sketch $y[n] = x[n] * h[n]$

- $x[n] = \{\underset{\uparrow}{1}, 1, 1\}$ $\quad h[n] = [1, 2, \underset{\uparrow}{2}, 1]$

1 Lesson Objectives

2 Properties

3 Homework

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3 Homework

# Lesson Objectives

At the end of this lesson, you should be able to

# Lesson Objectives

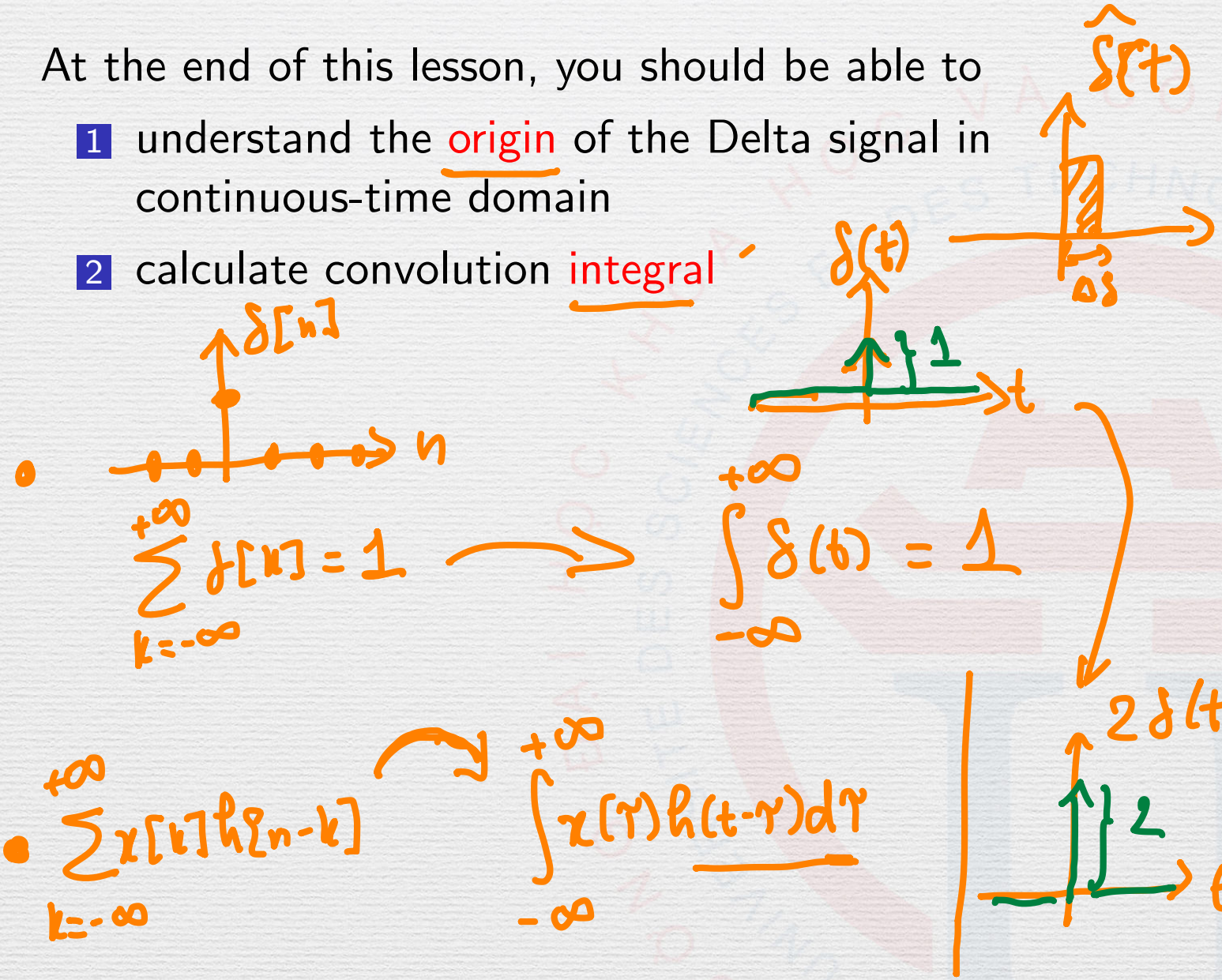
At the end of this lesson, you should be able to

1. understand the origin of the Delta signal in continuous-time domain

# Lesson Objectives

At the end of this lesson, you should be able to

1. understand the origin of the Delta signal in continuous-time domain
2. calculate convolution integral



1 Lesson Objectives

2 Properties

- Commutative
- Distributive
- Associative
- Causality
- Stability

3 Homework

# Commutative

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Objectives

Properties

Commutative

Distributive

Associative

Causality

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Homework

## Definition

$$x(t) * h(t) = h(t) * x(t)$$

Verify this property.

$$\int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = -\int_{+\infty}^{-\infty} x(t-\varphi)h(\varphi)d\varphi$$

$$\varphi = t - \tau \quad \hookrightarrow d\varphi = -d\tau$$

$$= \int_{-\infty}^{+\infty} x(t-\varphi)h(\varphi)d\varphi$$

$$= h(t) * x(t)$$

# Distributive

## Definition

$$x(t) * (h_1(t) + h_2(t)) = x(t) * h_1(t) + x(t) * h_2(t)$$

$h(t)$

Verify this property.

$$\int_{-\infty}^{+\infty} x(\tau)\,[h_1(t-\tau)+h_2(t-\tau)]\,d\tau = \int_{-\infty}^{+\infty} [x(\tau)h_1(t-\tau)+x(\tau)h_2(t-\tau)]\,d\tau$$

$$= \int_{-\infty}^{+\infty} x(\tau)h_1(t-\tau)\,d\tau + \int_{-\infty}^{+\infty} x(\tau)h_2(t-\tau)\,d\tau$$

$$= x(t) * h_1(t) + x(t) * h_2(t)$$

# Distributive - Parallel Interconnection

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Objectives

Properties

Commutative

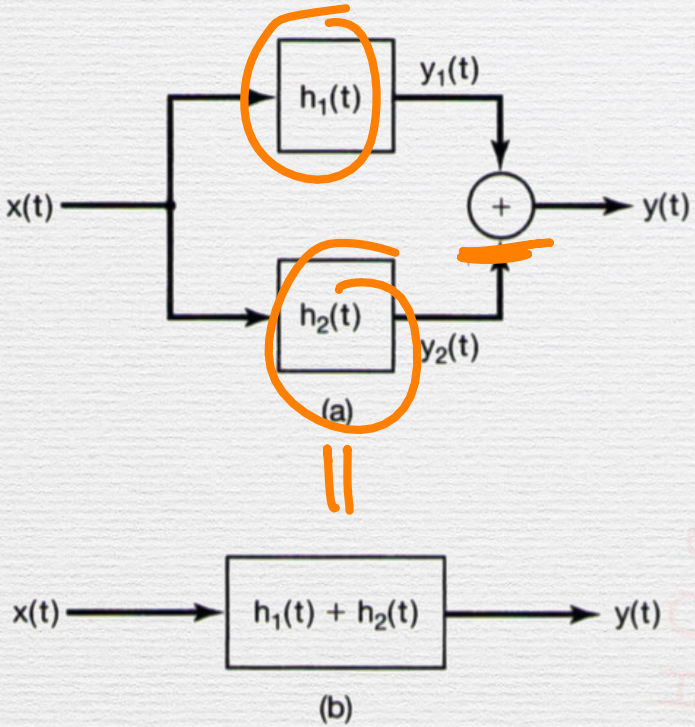
Distributive

Associative

Causality

Stability

Homework




$x(t) * h_1(t) + x(t) * h_2(t)$

$=$

$x(t) * (h_1(t) + h_2(t))$

# Associative

## Definition

$$x(t) * (h_1(t) * h_2(t)) = (x(t) * h_1(t)) * h_2(t)$$

$h(t)$

Verify this property.

- $h(t) = \int_{-\infty}^{+\infty} h_1(\tau)\, h_2(t-\tau)\, d\tau$

- $x(t) * (h_1(t) * h_2(t)) = \int_{-\infty}^{+\infty} x(\varphi) \left[ \int_{-\infty}^{+\infty} h_1(\tau)\, h_2(t-\varphi-\tau)\, d\tau \right] d\varphi$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(\varphi)\, h_1(\tau)\, h_2(t-\varphi-\tau)\, d\tau\, d\varphi$$

$$= \int_{-\infty}^{+\infty} h_1(\tau) \left[ \int_{-\infty}^{+\infty} x(\varphi)\, h_2(t-\varphi-\tau)\, d\varphi \right] d\tau$$

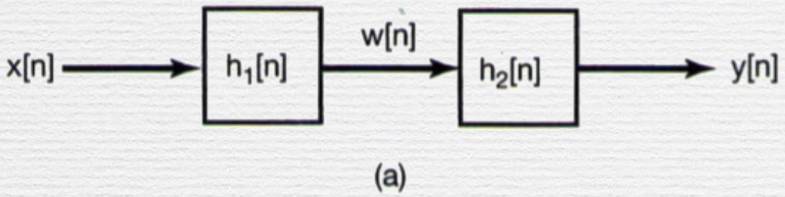
$= h_1(t) * (x(t) * h_2(t))$

- $x(t) * (h_1(t) * h_2(t)) = x(t) * (h_2(t) * h_1(t))$

$= h_2(t) * (x(t) * h_1(t))$

$x(t) * (h_1(t) * h_2(t)) = (x(t) * h_1(t)) * h_2(t)$

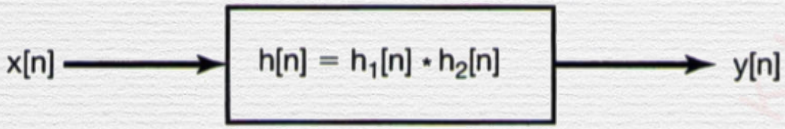
# Associative - Series Interconnection




(a)

$$(x(t) * h_1(t)) * h_2(t)$$

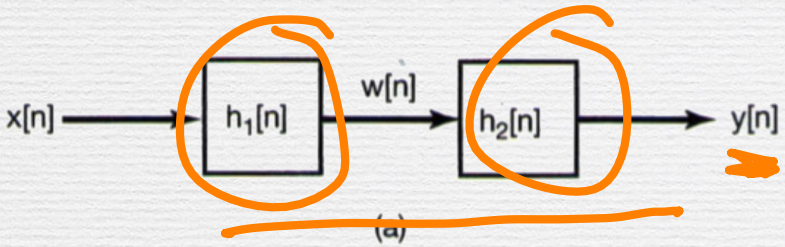
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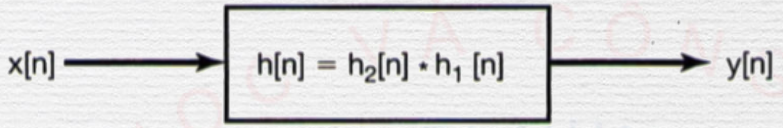

(b)

$$x(t) * (h_1(t) * h_2(t))$$

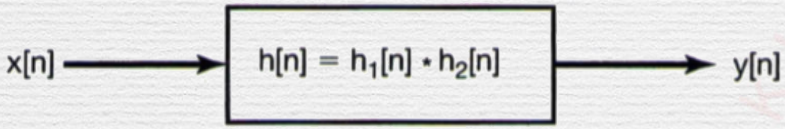
# Associative & Commutative



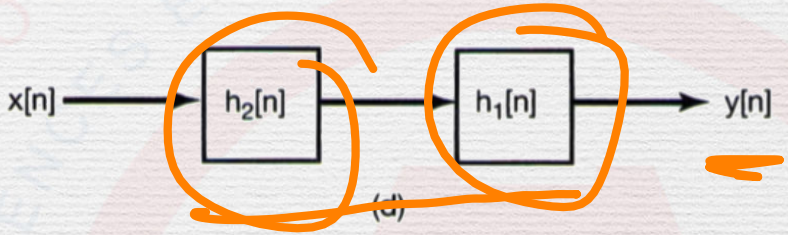

(a)




(c)




(b)




(d)

# Causality

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Objectives

Properties

Commutative

Distributive

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Stability

Homework

A LTI system is causal iff its impulse response satisfy the condition:

$$h(t) = 0 \text{ for } t < 0$$

# Stability

Signals & Systems

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Homework

A LTI system is stable iff its impulse response is absolutely integrable

$$\int_{-\infty}^{+\infty} |h(t)|dt < \infty$$

$$\sum_{k=-\infty}^{+\infty} |h[k]| < \infty$$

1 Lesson Objectives

2 Properties

3 Homework

# Homework

## Continuous LTI Systems Exercises

2.8, 2.9, 2.10, 2.22, 2.23

$x(t) * \delta(t) = x(t)$

$x(t) * \delta(t - t_0) = x(t - t_0)$

$$x(t) = \begin{cases} t+1, & 0 \le t \le 1 \\ 2-t, & 1 < t \le 2 \\ 0, & \text{else} \end{cases}$$

$$x(t) * h(t) = x(t) * (\delta(t+2) + 2\delta(t+1))$$

$$= x(t) * \delta(t+2) + 2x(t) * \delta(t+1)$$

$$= x(t+2) + 2x(t+1)$$

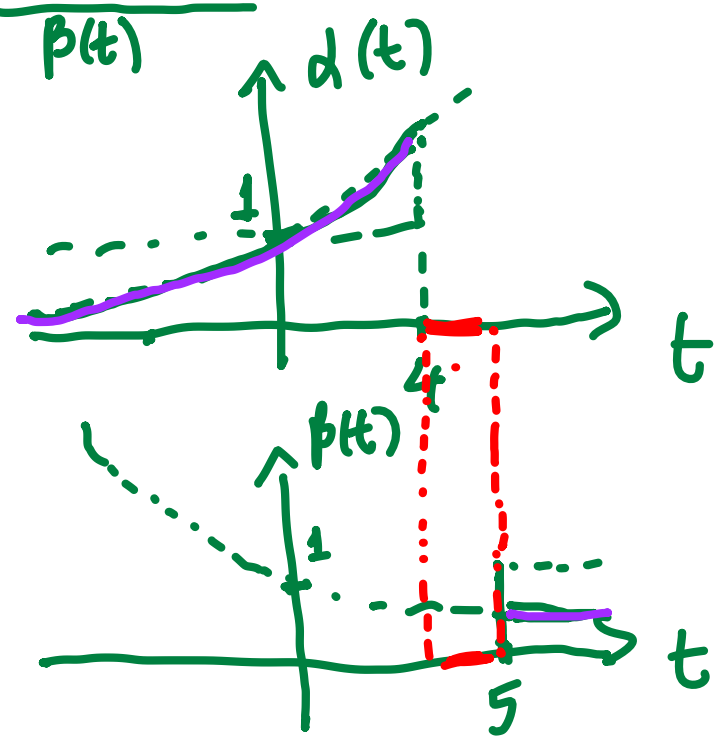
$$x(t+2) = \begin{cases} t+3, & -2 \le t \le -1 \\ -t, & -1 < t \le 0 \\ 0, & \text{else} \end{cases}$$

$$x(t+1) = \begin{cases} t+2, & -1 \le t \le 0 \\ 1-t, & 0 < t \le 1 \\ 0, & \text{else} \end{cases}$$

$$x(t) = x(t+2) + 2x(t+1) = \begin{cases} t+3, & -2 \le t \le -1 \\ t+4, & -1 < t \le 0 \\ 2-2t, & 0 < t \le 1 \\ 0, & \text{else} \end{cases}$$

2.9) $h(t) = \underbrace{e^{2t}u(-t+4)}_{\alpha(t)} + \underbrace{e^{-2t}u(t-5)}_{\beta(t)}$

$$h(t) = \begin{cases} e^{2t}, & t < 4 \\ e^{-2t}, & t > 5 \\ 0, & 4 < t < 5 \end{cases}$$

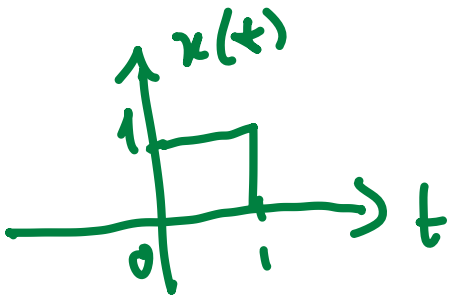


$$h(t-\tau) = \begin{cases} e^{2(t-\tau)} & , \quad t-\tau < 4 \\ e^{-2(t-\tau)} & , \quad t-\tau > 5 \\ 0 & , \quad 4 < t-\tau < 5 \end{cases}$$

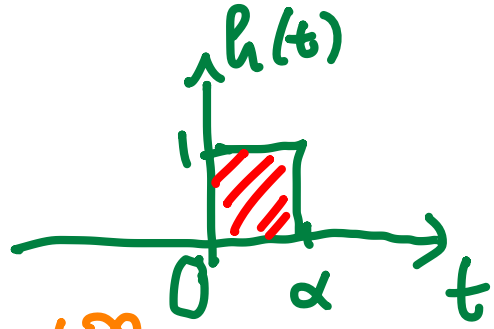
$$\begin{cases} \tau > \underbrace{t-4}_{B} \\ \tau < \underbrace{t-5}_{A} \end{cases}$$

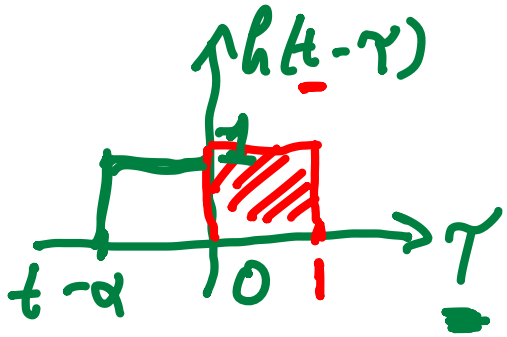
$$\underbrace{t-5}_{A} < t < \underbrace{t-4}_{B}$$

Conclusion: $\begin{cases} A = t-5 \\ B = t-4 \end{cases}$



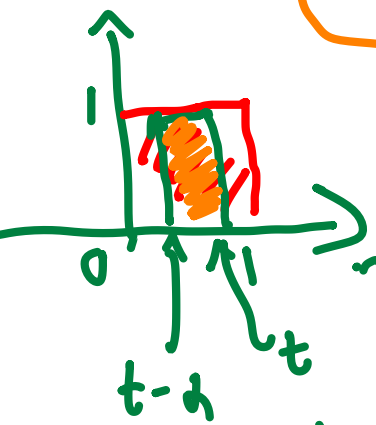

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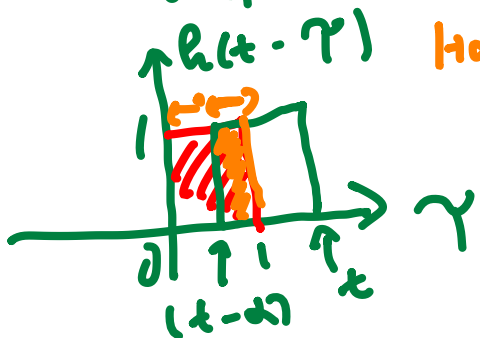




$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)\, h(t-\tau)\, d\tau .$$

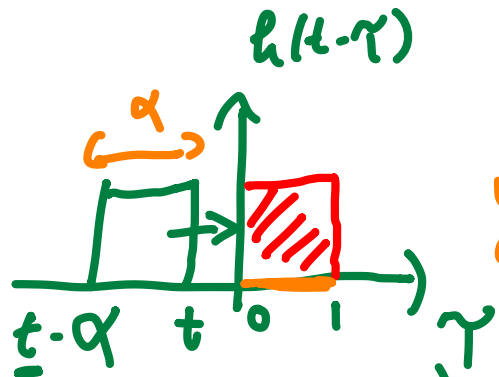
$$= \int_{0}^{1} h(t-\tau)\, d\tau$$



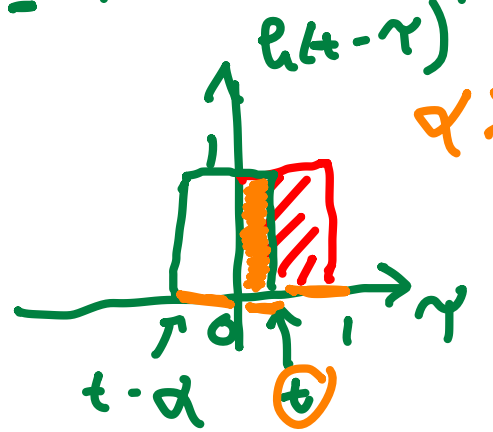

$0 < t \leq \alpha$, $\quad y(t) = t$



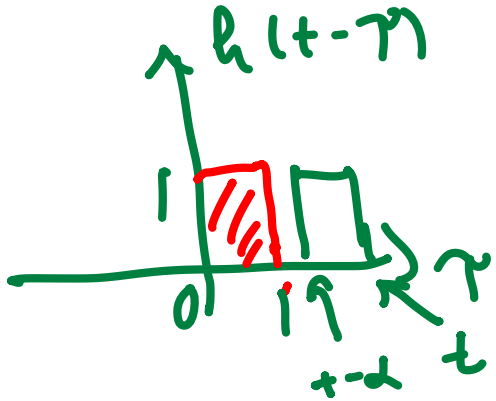

$t \geq \alpha$, $\quad y(t) = 1 - t + \alpha$




$t < 0$, $\quad y(t) = 0$




$\alpha \geq t > 0$, $\quad y(t) = t$




$t > 1+\alpha$,

$y(t) = 0$

Try to solve this at home. $\Rightarrow$

